

Network Management: DHCP & DNS

1.0 DHCP (Application Layer Protocol)

Dynamic Host Configuration Protocol (DHCP). Seamless network connectivity is achieved by automatically assigning IP addresses and configuration settings to devices using the Dynamic Host Configuration Protocol (DHCP).

- Assigns a unique IP address to each device.
- Provides the subnet mask to define the local network range.
- Configures the default gateway for communication outside the local network.
- Supplies DNS server addresses for domain name resolution.

DHCP Configuration lifecycle (DORA):

D - Discover

O - Offer

R - Request

A - Acknowledgement

In this lab, we will do the following:

- Install the Kea DHCP Server (on our server machine).
- Configure Kea to hand out IPv4 addresses for the 192.168.3.0/24 subnet
- Reserve 192.168.3.90/24 for the client machine
- Lease time: 6 hours
- Other DHCP options: DNS Servers

Configure ens19 of our server with static IP

Command:

```
nmcli connection modify ens19 ipv4.addresses 192.168.3.1/24 ipv4.method manual
connection.autoconnect yes
```

```
nmcli connection up ens19
```

The above command says, whenever system starts/restarts and ens19 exists, activate this config.

```
root@NM-G10-H1:~# nmcli connection modify ens19 ipv4.addresses 192.168.3.1/24 ipv4.method manual connection.autoconnect yes
root@NM-G10-H1:~# nmcli connection up ens19
connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/31130)
root@NM-G10-H1:~# ip a
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:14:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0c29
    altname enx0202000a0001
    inet 172.17.10.1/25 brd 172.17.10.127 scope global dynamic noprefixroute ens18
        valid_lft 66605sec preferred_lft 66605sec
    inet6 fe80::9abc:82b3:9f69:624d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ens19: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:14:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0c29
    altname enx0202010a0001
    inet 192.168.3.1/24 brd 192.168.3.255 scope global noprefixroute ens19
        valid_lft forever preferred_lft forever
    inet6 fe80::113c:717f:a849:5f38/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@NM-G10-H1:~#
```

Configure Kea DHCP

```
vi /etc/kea/kea-dhcp4.conf
```

```
{
    "Dhcp4": {
        "interfaces-config": {
            "interfaces": [ "ens19" ]
        },

        "lease-database": {
            "type": "memfile",
            "persist": true,
            "name": "/var/lib/kea/kea-leases4.csv"
        }
    }
}
```

},

"renew-timer": 3600,

"rebind-timer": 7200,

"valid-lifetime": 21600,

"subnet4": [

{

"subnet": "192.168.3.0/24",

"pools": [

{

"pool": "192.168.3.150 - 192.168.3.200"

}

],

"option-data": [

{

"name": "domain-name-servers",

"data": "192.168.3.1"

}

],

"reservations": [

{

"hw-address": "CLIENT_MAC_ADDRESS",

"ip-address": "192.168.3.90"

}

]

}

]

```
}  
}
```

Start Kea DHCP Server

systemctl start kea-dhcp4-server

systemctl status kea-dhcp4-server

systemctl enable kea-dhcp4-server (persist through system reboot)

```
root@NM-G10-H1:~# vi /etc/kea/kea-dhcp4.conf
root@NM-G10-H1:~# systemctl start kea-dhcp4-server
root@NM-G10-H1:~# systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-05-08 18:10:34 CEST; 3s ago
     Invocation: 2f4815cf3bc24f1691a1596727cd6433
       Docs: man:kea-dhcp4(8)
    Main PID: 26148 (kea-dhcp4)
      Tasks: 7 (limit: 2300)
    Memory: 2.9M (peak: 3.1M)
       CPU: 39ms
    CGroup: /system.slice/kea-dhcp4-server.service
           └─26148 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.870 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface ens19
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.870 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type" not specified, using default socket type
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.870 INFO [kea-dhcp4.dhcp4/26148.140540373096384] DHCP4_CONFIG_COMPLETE DHCPv4 server has completed configuration: added IPv4 subnets: 1; DDNS: disabled
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.873 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_MEMFILE_DB opening memory file lease database: lfc-interval=3600 name=/var/lib/kea/kea-leases4.csv
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.874 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_MEMFILE_LEASE_FILE_LOAD loading leases from file /var/lib/kea/kea-leases4.csv
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.874 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_MEMFILE_EXTRACT_EXTENDED_INFO saw 0 leases, extended info saw 0 leases
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.874 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_MEMFILE_LFC_SETUP setting up the Lease File Cleanup interval to 3600 sec
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.896 INFO [kea-dhcp4.dhcpsrv/26148.140540373096384] DHCP4_SRV_CFGMGR_USE_ALLOCATOR using the iterative allocator for V4 leases in subnet 192.168.3.1
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.896 WARN [kea-dhcp4.dhcp4/26148.140540373096384] DHCP4_MULTI_THREADING_INFO enabled: yes, number of threads: 2, queue size: 64
May 08 18:10:34 NM-G10-H1 kea-dhcp4[26148]: 2026-05-08 18:10:34.896 INFO [kea-dhcp4.dhcp4/26148.140540373096384] DHCP4_STARTED Kea DHCPv4 server version 2.6.3 started

Synchronizing state of kea-dhcp4-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable kea-dhcp4-server
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp4-server.service' → '/usr/lib/systemd/system/kea-dhcp4-server.service'.
root@NM-G10-H1:~# systemctl is-enabled kea-dhcp4-server
enabled
root@NM-G10-H1:~#
```

You can see the ipv4 address handed out by the DHCP server

cat /var/lib/kea/kea-leases4.csv

```
root@NM-G10-H1:~# ssh root@192.168.3.90
root@192.168.3.90's password:
Linux localhost.localdomain 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

Last login: Sat May  9 18:02:29 2026 from 192.168.3.1

#####
# vLab 2.0
#
# IPv4 address: 192.168.3.90
# Note: This machine is not connected to a vLab team access network
#####
```

If not the reserved address, you can force client DHCP renew or reboot client:

nmcli conn down ens18 && nmcli conn up ens18

```
root@localhost:~# nmcli conn down ens18 && nmcli conn up ens18
Connection 'ens18' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/31185)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/31186)
root@localhost:~# cat /etc/resolv.conf
# Generated by NetworkManager
search mydomain.example.com example.com
nameserver 192.168.3.1
root@localhost:~#
```

2.0 DNS

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex, newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).

Configure Forwarding

```
vi /etc/bind/named.conf.options
```

```
options {  
    directory "/var/cache/bind";  
  
    forwarders {  
        172.17.10.126;  
    };  
  
    forward only;  
  
    dnssec-validation no;  
  
    listen-on { any; };  
    allow-query { any; };  
};
```

- forwarders — sends all unknown DNS requests to your vLab DNS server
- forward only — don't try to resolve itself, always forward
- dnssec-validation no — avoids validation issues in the lab environment

- allow-query { any; } — allows the client to query this DNS server

1. Make Bind authoritative for the domain [g10.nm.vlab....](#)
2. Resolve the IP address of the server (the one on ens19!) to `server.g<group>.nm.vlab...`
3. Resolve the IP address of the client to `client.g<group>.nm.vlab....`
4. Always resolve `www.<x>.at` to [19x.xxx.127.228](#).

```
root@NM-G10-H1:/etc/bind# cat named.conf.local

//
// Do any local configuration here
//
zone "g10.nm.vlab." {
    type master;

    file "/etc/bind/db.g10.nm.vlab.";
};
zone "jku.at" {
    type master;

    file "/etc/bind/db.jku.at";
};
```

```
root@NM-G10-H1:/etc/bind# cat db[redacted].at
$TTL 86400
IN SOA server.g10.nm.vlab.[redacted].at. admin.g10.nm.vlab.[redacted].at. (
2024010101
3600
1800
604800
86400 )
```

```
@ IN NS server.g10.nm.vlab.ins.jku.at.
server.g10.nm.vlab.[redacted].at IN A 192.168.3.1
www IN A 192.168.3.90
```

```
root@NM-G10-H1:/etc/bind# cat db.g10.nm.vlab.[redacted].at
$TTL 86400
@ IN SOA server.g10.nm.vlab.[redacted].at. admin.g10.nm.vlab.[redacted].at. (
2024010101 ; Serial
3600 ; Refresh
1800 ; Retry
604800 ; Expire
86400 ) ; Negative TTL
```

```
; Name servers
```

```
@ IN NS server.g10.nm.vlab.[redacted].at.
```

```
; A records
```

```
server IN A 192.168.3.1
client IN A 192.168.3.90
```

```
root@NM-G10-H1:/etc/bind# cat named.conf.options
options {
```

```
    directory "/var/cache/bind";
    forwarders {
        172.17.10.126;
    };
    forward only;
    dnssec-validation no;
    listen-on { any; };
    allow-query { any; };
};
```

```
root@NM-G10-H1:/etc/bind# cat named.conf.local
```



```
root@localhost:~# dig server.g10.nm.vlab.192.168.3.1

; <<>> DiG 9.20.18-1~deb13u1-Debian <<>> server.g10.nm.vlab.192.168.3.1
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21335
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 08a44916da7f9d44010000006a0068a9b3bd5752da1a53e5 (good)
;; QUESTION SECTION:
;server.g10.nm.vlab.192.168.3.1. IN      A

;; ANSWER SECTION:
server.g10.nm.vlab.192.168.3.1. 86400 IN A      192.168.3.1

;; Query time: 0 msec
;; SERVER: 192.168.3.1#53(192.168.3.1) (UDP)
;; WHEN: Sun May 10 13:15:16 CEST 2026
;; MSG SIZE rcvd: 102
```

```
root@localhost:~# dig client.g10.nm.vlab.192.168.3.90

; <<>> DiG 9.20.18-1~deb13u1-Debian <<>> client.g10.nm.vlab.192.168.3.90
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29155
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: c0a56f5658ff834f010000006a0068b4e7be2536bb396bae (good)
;; QUESTION SECTION:
;client.g10.nm.vlab.192.168.3.90. IN      A

;; ANSWER SECTION:
client.g10.nm.vlab.192.168.3.90. 86400 IN A      192.168.3.90

;; Query time: 0 msec
;; SERVER: 192.168.3.1#53(192.168.3.1) (UDP)
;; WHEN: Sun May 10 13:15:27 CEST 2026
;; MSG SIZE rcvd: 102
```


Reference

- <https://www.geeksforgeeks.org/computer-networks/dynamic-host-configuration-protocol-dhcp/>
- <https://www.cloudflare.com/learning/dns/what-is-dns/>